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ERP FAILURE: A SYSTEMATIC MAPPING OF THE LITERATURE

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Abstract:

With the development of different technologies, the use and importance of Enterprise Resource Planning (ERP) systems continue to increase daily. In parallel with this increasing use, a lot of research is being done to successfully complete ERP implementation projects. However, despite these researches, reported case studies show that the success rates of ERP projects are meager.

Based on the examples of experienced failure, researchers determine very different failure factors with varying perspectives for companies in different industry sectors, cultures, and sizes. It is becoming increasingly difficult for many practitioners and researchers to understand these failure factors correctly. Our objective is to investigate the state-of-the-art ERP Failure Factors that could benefit practitioners to utilize that information potentially.

We review the body of knowledge related to ERP failure factors in the form of a systematic literature mapping (SLM). We pose four sets of research questions and systematically develop and refine a classification schema. The initial pool consisted of 353 articles. Systematic voting was conducted among the authors regarding the inclusion/exclusion criteria. As a result, there were 72 technical articles in our final pool.

This SLM provides an overview of ERP critical failure factors (CFF) with different focused headings. These headings cover qualitative coding about CFF names, CFF rankings, the relation between CFF and ERP processes and failure modes, etc.

The results of this study would benefit three groups of stakeholders: *i)* Researchers who work on ERP Failure Factors, *ii)* Solution implementers who provide consultancy services to companies that carry out ERP Implementation projects, and *iii)* ERP project implementation managers. These stakeholders could utilize the results of this SLM to catch the trend of ERP implementation challenges.

Keywords: ERP, enterprise resource planning, failure; failure factor; systematic literature mapping

1 INTRODUCTION

Efficient information flow is significant in terms of business management in organizations. In other words, for an organization to work correctly, holistic, fast, and reliable information is required in production, human resources, purchasing, financial affairs, and stock [1]. For this purpose, management information systems should be implemented efficiently. In this context, ERP is the general name for systems and software developed to ensure or support the end-to-end management and efficient use of enterprises by bringing together human resources, physical resources, and financial resources [2]. If an organization wants to gain a competitive advantage by increasing its productivity, its ERP system should be implemented successfully.

There are many examples where ERP has been successfully implemented. However, understanding and implementing ERP involves high risk and costs. There are many difficulties in the implementation phase of an ERP system because it affects processes, employees, and even culture in an organization [3]. Moreover, organizations should change their already established systems and restructure and improve their business processes [4]. Another critical challenge is that ERP must align with the strategic plan [5]. Apart from these, many other difficulties need to be taken into consideration. With all this in mind, no doubt that implementing a successful ERP system is a very challenging task.

A lot of research has been done to identify and understand the challenges of ERP Implementation [6]. To handle the challenges of ERP Implementation, different criteria have been determined according to the organization's field of activity, size, and location. Moreover, approaches, models, and frameworks have been created for a successful ERP implementation.

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As the ERP market grows globally, research on a successful ERP implementation is also increasing. Despite all these researches and defined factors, the success of ERP projects is still not sure.

Despite all efforts, recent research shows that a high failure rate continues in ERP implementation projects [7], [8]. At the same time, from such a broad perspective, it is tough for a new researcher and practitioner to review the ERP failure factors from all aspects. From the authors' view, lessons learned from failed projects are more critical than how success can be achieved. It is essential to correctly determine, analyze, and understand the failure criteria in ERP implementation. Consideration of various failure factors in the execution of the ERP projects may benefit organizations, consultants, and ERP vendors. However, finding and reading all related research in this field will be a challenging task to benefit from the existing knowledge in this field. Therefore, we observed that there is a need for a comprehensive mapping paper that provides an "index" of knowledge in this area and summarizes the field from the perspective of "why did they fail" rather than "how to be successful".

Considering the failure rate in ERP projects, we carry out this research in the systematic literature mapping approach to meet the above need. Our review pool consists of 72 academic peer-reviewed articles. Our mapping study covers all published articles from 1999 to the end of 2020. In ERP implementation projects, while there are many review articles from the "how to be successful" perspective, there are only a few articles from the "why did they fail" attitude. As we will give in detail in Section 2.3, considering all the 72 articles in our final pool, our study is the most up-to-date and comprehensive mapping study done in this area so far.

The paper is divided into the following remaining sections. Section 2 presents background and related work. The research method is described in Section 3. The results of our mapping study are presented in Section 4. Section 5 presents the findings of this review. Lastly, conclusions and future work are provided in Section 6. Additionally, the list of the reviewed studies in this survey can be seen in Appendix 1.

2 Background And Related WORK

In this section, firstly, a brief overview of ERP is provided and then followed by an overview of ERP failure. We then provide the related works, which are the existing review papers on ERP Failure factors.

2.1. What Is ERP?

ERP was first called in 1990 by The Gartner Group to describe the next generation of Material Requirements Planning (MRP) software. It aimed to integrate all business processes under one application [9]. ERP system is designed to integrate and make better the business processes and transactions in a corporation [10]. Later, the definition of ERP would be expanded to include nearly any large integrated software package [11]. However, in order to integrate business processes, organizations should re-engineer their business routines to match routines embedded in ERP systems [12]. In this way, an ERP system helps organizations to achieve and manage their business routines efficiently and effectively [13]. Moreover, cross-functional integration was a new concept for many organizations during the 1990s and 2000s [14], however, after those years, cross-functional operations became increasingly common for ERP. Gathering all information from the ERP system ensures easy and fast access to the required information by eliminating the duplicate entry of information. Thus, information management and information security can be realized much more easily and reliably.

There is no doubt that the implementation of ERP systems is different from other systems [15]. Although ERP is used to manage the processes of organizations, it is more efficient than traditional systems. One of the reasons that distinguishes the ERP system from other software is that it offers a successful ROI. Thanks to the central database, it provides improved collaboration between all departments within the organization. Thanks to this "improved collaboration", ERP increases the analytics and reporting capabilities of the organization. In addition, ERP provides an advanced decision support infrastructure. In recent years, customer management has become very important, and happy customers may be possible thanks to ERP. Moreover, ERP increases productivity by allowing you to automate ordinary tasks. ERP also offers integrated solutions for inventory management.

The main reason why organizations prefer ERP systems is the integration of business units with each other. This can improve the competitive position of the organization [16]. According to a survey conducted in 2017, 17% of organizations are implementing ERP to improve business performance. According to the same research, 14% of organizations implement ERP systems for ensure compliance, 14% for make employees' jobs easier, and 13% better integrate systems across locations [17].

The use of ERP, which initially started in large-scale companies, is becoming widespread in small and medium-sized companies too. With the developing technology, there is no doubt that the use of ERP will continue to increase day by day. The Factory of the Future (FoF) is based on the technological concepts of cyber-physical systems, Internet of Things (IoT). With next-generation ERP systems, within the modular structured of smart factories, in order to make decentralized decisions, cyber systems monitor the processes and create a virtual copy. Over the IoT, cyber-physical systems cooperate with not only humans but also each other in real time [18]. Given all these developments, the ERP applications market size was $94 billion in 2019, and according to estimates, it will approach $ 100 billion in 2024 [19].

The promises of ERP vendors continue to be attractive to organizations. However, it is uncertain how much ERP vendors and solution implementers will stick to their promises because of the high failure rate encountered in ERP implementation. We will examine this failure in the following sections.

2.2. Critical Failure Factors of ERP Systems

It is a fact that ERP systems are extremely important for organizations to survive in the competitive world. However, it is not an easy task to implement the ERP system successfully. The more expensive it is to implement ERP systems, the greater the negative financial impact of unsuccessful implementations. It is stated that many companies suffer from failed ERP system implementation [20]. In 2004, it was evaluated that 90% of ERP applications were delivered late or exceeded the budget, and 67% of these applications failed to achieve expected benefits realization [21]. According to a report published in 2017, only 26% of the organizations stated that they were satisfied with their ERP vendors [17].

ERP vendors are constantly telling their success stories on their web pages and in the magazines they publish. However, it is very important to analyze unsuccessful case studies and learn lessons from these cases in order to achieve success. There are enormous case studies in which the failures in ERP applications and the losses caused by these failures are explained with all clarity. For example, Nike lost $ 100 million in sales due to the wrong ERP selection and faulty supply chain design. Another example is for Hewlett Packard. Hewlett Packard failed ERP implementation due to bad scope planning and implementation with a big bang approach rather than a phased approach. On the other hand, Ingram Micro suffered great losses due to operational issues resulting from ERP implementation [22].

To be able to examine failures in ERP implementation, it is necessary to fully understand the definition of failure. Most of the organizations do not select the vendor & software again. One of the most common types of failure in ERP implementation can be seen as budget overruns. Often, budget overruns are caused by issues such as expanded scope, additional technology requirements, unanticipated organizational issues. Many ERP implementations exceeded their initial project timeline because of issues such as expanded scope and data issues. Moreover, another type of failure is that organizations are not getting the expected benefits from ERP systems.

Much of the research done in ERP implementation uses various theoretical frameworks [23]. However, there are studies in the literature on evaluation criteria for the failure of ERP applications [24]. Lack of management support, poor organizational culture, communication problems, financial management problems, lack of clarity on scope, project management problems, poor project team, ERP software misfit, poor solution implementer performance, poor quality of testing, outdated it infrastructure, lack of training, highly customized ERP, poor process design, lack of user involvement, resistance to change, unrealistic expectations are the main failure factors that need to be analyzed in detail.

2.3. Review Of Secondary Studies Regarding ERP

The work we have done is a mapping of primary studies. Therefore, it is a secondary study in the field of ERP Failure. However, there is a lot of secondary work on ERP, especially related to success factors. Therefore, it will be useful to briefly mention important secondary studies on ERP. Table 1 shows a list of key secondary studies in ERP.

Paper Title	Publication Year	Authors	Place of Publication	Reference
Enterprise resource planning: an integrative review	2004	E. M. Shehab, M. W. Sharp, L. Supramaniam, and T. A. Spedding	Business Process Management Journal	[25]

A survey on the recent research literature on ERP systems	2005	V. Botta-Genoulaz, P. A. Millet, and B. Grabot	Computers in Industry	[26]
Enterprise Resource Planning (ERP): a review of the literature	2007	Y. Moon	International Journal of Management and Enterprise Development	[10]
ERP implementation: a compilation and analysis of critical success factors	2007	S. Finney and M. Corbett	Business Process Management Journal	[27]
Examining the critical success factors in the adoption of enterprise resource planning	2008	E. W. T. Ngai, C. C. H. Law, and F. K. T. Wat	Computers in Industry	[28]
A review of critical success factors for ERP projects	2009	S. A. Kronbichler, H. Ostermann, and R. Staudinger	The Open Information Systems Journal	[29]
Manufacturing execution system - a literature review	2009	B. S. De Ugarte, A. Artiba, and R. Pellerin	Production Planning and Control	[30]
Successful enterprise resource planning implementation taxonomy of critical factors	2009	S. Dezdar and A. Sulaiman	Industrial Management & Data Systems	[31]
A comprehensive literature review of the ERP research field over a decade	2010	B. R. Schlichter and P. Kraemmergaard	Journal of Enterprise Information Management	[32]
A review of ERP research: a future agenda for accounting information systems	2011	S.V. Grabski, S. A. Leech, P. J. Schmidt	Journal of Information Systems	[33]
Critical success factors in enterprise resource planning systems: review of the last decade	2013	L. Shaul and D. Tauber	ACM Computing Surveys	[34]

Table 1. Key secondary studies in ERP

Studies on "success factors" prepared with the "how to be successful" approach are beyond the scope of this research. However, while conducting a secondary study on the "Failure Factor" prepared with the "why did they fail" approach, the studies presented in Table 1 formed the basis. For example, Shaul [34] only focused on Success Factors for implementing ERP. Dezdar [31] presented a systematic compilation of success factors and created a comprehensive taxonomy of them. Schlichter [32] presented a methodological framework for doing an extensive literature review. Botta [26] analyzed the literature according to the following categories: ERP implementation, ERP optimization, ERP management, ERP software, and supply chain. Shehab [25] presented a review of the ERP-focused studies between 1990-2003. Finney [27] reviewed the literature base of critical success factors to explore any gaps that might exist. Ngai [28] provided a literature review of the critical success factors in ERP implementation across 10 different regions. Ugarte [30] provided a literature review of manufacturing execution system (MES) relationships with other ERP systems.

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2.4. Related Works: Review Studies Regarding ERP Failure

A comprehensive secondary study focused solely on failure factors has not yet been found in the literature. Many of the studies address the challenges encountered during the implementation of the ERP system. We were able to find 6 such studies [3, 35, 36, 37, 38, 39] and show their list in Table 2. For each related work, we include the publication year, its type (a conventional survey or systematic literature review – SLR), several reviewed papers in the pool, and some explanatory notes.

Paper Title	Publication Year	Authors	Place of Publication	Type of Review	# of Papers in The Pool	Subgroups and Notes	Reference
Literature review on ERP implementation challenges	2016	S. Ranjan, V. K. Jha, and P. Pal	International Journal of Business Information Systems	A conventional survey	27	**ERP implementation challenges:** The article repository was not focused on failure. The repository consists of articles on the implementation challenges.	[3]
Benefits and challenges of cloud ERP systems–a systematic literature review	2016	M. A. Abd Elmonem, E. S. Nasr, and M. H. Geith	Future Computing and Informatics Journal	SLR	31	**Challenges related to cloud ERP systems:** The article repository was not focused on failure. The repository consists of articles on the challenges in cloud ERP Systems.	[35]
Identification of key success factors and challenges for ERP systems—a systematic literature review	2017	S. F. Wijaya, Meyliana, H. Prabowo, and R. Kosala	International Conference on Applied Computer and Communication Technologies (ComCom)	SLR	29	**ERP implementation challenges:** The article repository was not focused on failure. The repository consists of articles on the implementation challenges.	[36]
ERP systems in multinational enterprises: a literature review of post-implementation challenges	2018	K. B. Osnes, J. R. Olsen, P. Vassilakopoulou, and E. Hustad	Procedia Computer Science	SLR	20	**Challenges related to post-ERP implementation:** The article repository was not focused on failure. The repository consists of articles on the post-implementation challenges.	[37]
Factors influencing enterprise resource planning system: a review	2018	M. N. . Hasan, M.S., Ebrahim, Z., Mahmood, W.H.W. and Rahman	Journal of Advanced Manufacturing Technology	A conventional survey	19	**Factors affecting ERP:** The article reviews on factors that influence ERP between 2011 and 2016.	[38]
ERP issues and challenges: a research synthesis	2019	F. Mahmood, A. Z. Khan, and R. H. Bokhari	Kybernetes	SLR	53	**ERP implementation challenges:** The repository consists of articles on the implementation challenges.	[39]
Our study	2021	-	-	SLM	72	We review the body of knowledge related to ERP failure factors in the form of a systematic literature mapping (SLM). We pose four sets of research questions and systematically develop and refine a classification schema. The initial pool consisted of 353 articles. Systematic voting was conducted among the authors regarding the inclusion/exclusion criteria. As a result, there were 72 technical articles in our final pool.	

Table 2. Important secondary studies in ERP challenges

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3 RESEARCH METHOD

We performed a Systematic Literature Mapping (SLM) study methodology for this research work. This methodology proceeds step by step by following PICO (Population, Intervention, Comparison and Outcomes) search strategy suggested by Kitchenham and Charters [40] as shown below:

- Population: ERP Failure
- Intervention: ERP Failure Factors
- Comparison: Characteristics of ERP Failure Factors
- Outcomes: Models, attributes, metrics, and performance evaluations for ERP Failure Factors
- Context: Academia (scientific literature)

The objective of this approach is to extract all relevant literature studies in the context of ERP failure for answering defined research questions (RQs). To perform SM, we follow the guidelines recommended by Kitchenham [40] and Petersen et al. [41]. We follow six steps in order to conduct this SM: (i) research goal and questions; (ii) strategy for source selection; (iii) inclusion/exclusion criteria; (iv) data extraction; (v) results; and (vi) discussion.

3.1 RESEARCH GOAL AND QUESTIONS

In this SLM study, our objective is focusing on the following research goal: "To understand and define ERP failure factors". To answer this research goal, we defined research questions specific to our study, as described in Figure 1. As shown in Figure, we identified eighteen RQs under four headings.

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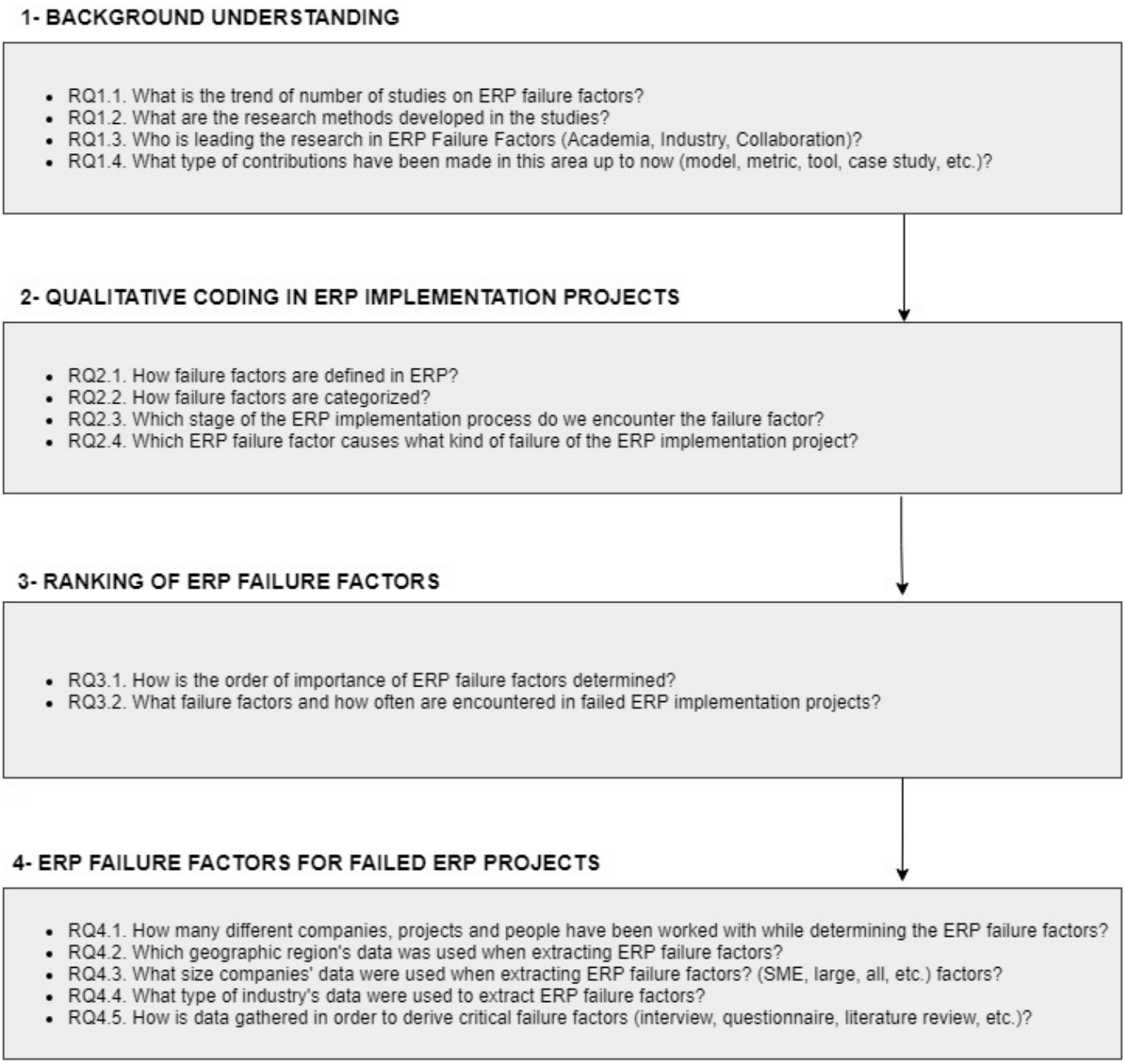


Figure 1. Research Questions identified in the SLM

3.2 Source Selection

In this SLM study, two source selection strategies which are string-based search and snowballing were followed to select appropriate literature studies. The visualization of our followed procedure specific to this study is shown in Figure 2.

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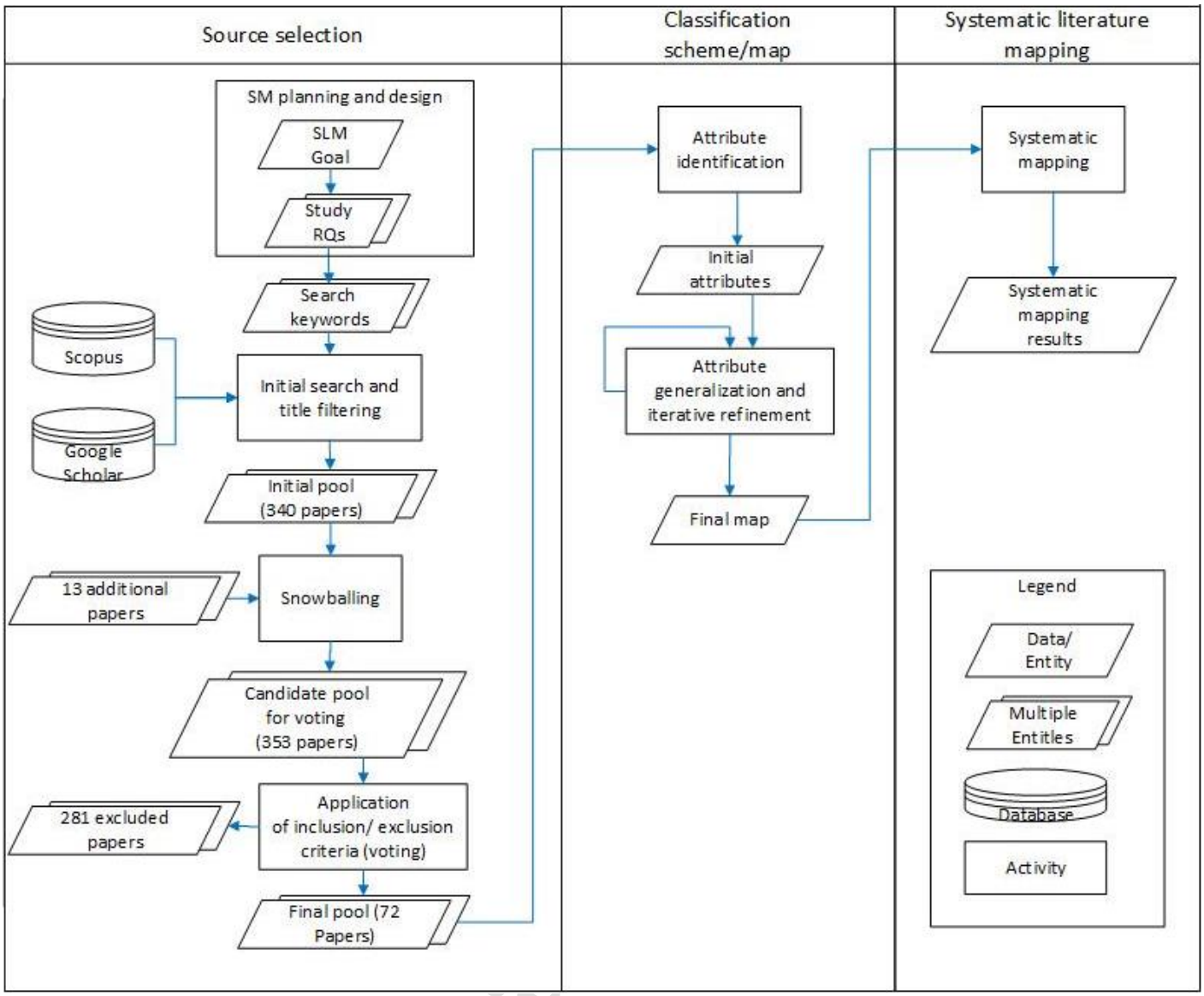


Figure 2. An overview of our research protocol

Firstly, we constructed our search key in order to perform a string-based search. Our search strings are shown in Table 3. By searching with these keywords, we look at the titles, keywords, and abstracts of papers in the literature, and we get relevant papers by searching primarily on Google scholar and Scopus database. We used keywords that were constructed on the date of April. 1, 2021 on two different databases. By using this search string, we searched in all databases separately, and listed the number of selected papers for each search string as shown in Table 3. By searching keywords by searching on the title, we found 340 relevant papers. In addition, it is possible that there may be some relevant studies that are missed by string-based search. In order to ensure not to miss any relevant studies from the initial search with string-based, we applied snowballing process that is recommended besides database searches in systematic mapping studies. By applying this technique, we found 13 additional articles and included them in our study pool, and we reached 353 relevant papers as an initial set. After applying inclusion/exclusion criteria to our candidate papers, we reached 72 papers as a final pool.

Search string	*Allintitle:* Enterprise resource planning AND fail *Allintitle:* Enterprise resource planning AND fails *Allintitle:* Enterprise resource planning AND failure *Allintitle:* Enterprise resource planning AND failures *Allintitle:* Enterprise resource planning AND unsuccessful *Allintitle:* Enterprise resource planning AND insufficient *Allintitle:* Enterprise resource planning AND disappoint *Allintitle:* Enterprise resource planning AND problem *Allintitle:* Enterprise resource planning AND problems *Allintitle:* Enterprise resource planning AND challenge *Allintitle:* Enterprise resource planning AND challenges *Allintitle:* ERP AND fail *Allintitle:* ERP AND fails *Allintitle:* ERP AND failure *Allintitle:* ERP AND failures *Allintitle:* ERP AND unsuccessful *Allintitle:* ERP AND insufficient *Allintitle:* ERP AND disappoint *Allintitle:* ERP AND problem *Allintitle:* ERP AND problems *Allintitle:* ERP AND challenge *Allintitle:* ERP AND challenges
Results quantity on Google Scholar	1002
Results quantity on Scopus	215

Table 3. Results quantity on each database

3.3 INCLUSION AND EXCLUSION CRITERIA

In order to assess each candidate primary study in systematic mapping studies, it is required to consider inclusion and exclusion criteria. We defined these criteria when the protocol was drafted. We select the potential primary studies by examining inclusion and exclusion criteria. Below, we described the inclusion and exclusion criteria that we followed during the SLM study.

Inclusion Criteria

Articles on the proposed topic, published until April 1st 2021, will be included if;

- Research paper focused on ERP Failure Factors
- Research paper includes a relatively sound validation
- The selected paper that is not the same or similar paper already in the pool

Exclusion Criteria

The following types of papers will be excluded if;

- The full text of the paper is not accessible
- The paper is not written in English
- The research paper is not relevant to ERP Failure
- The paper does not include a relatively sound validation
- The same or similar paper already in the pool

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- The research paper's type is Ph.D. dissertations, secondary studies (SLR/SLM), posters

In fact, the authors have faced with a certain degree of difficulty when performing the SLM study. Constructing research questions and deciding on search strategies were made by two authors. By considering inclusion/exclusion criteria, the final pool reached 72 articles from 340 initial papers. After preparing the final pool with included papers, we embedded all the articles into an Excel document and performed all of the data extraction processes on this document. In one iteration, two authors divided all the 72 articles into half, and they simultaneously extracted each of the 36 articles in order to find answers to the RQs. Then, each author checked the data extracted by the other author. If there is any conflict in this stage, the authors discussed the disagreement parts together, and proposed to make an agreement on these conflicts.

3.4 Classification Scheme/Map

In order to extract useful data from primary studies, we applied the data extraction process. For making an in-depth analysis, we performed this process on 72 articles in the final pool. Firstly, we gathered all the papers by downloading from the Google Scholar and Scopus databases. Then, after creating an Excel sheet, we recorded all of the required information into this sheet. In order to mitigate bias with the data extraction process, two authors conducted the data extraction process into two steps in which the first step is done by dividing all final pool into two, and each author extracted data simultaneously, and in the second step, they peer-reviewed. When there was an unsolved dispute between the two researchers, they first discussed the reasons that they support, and then they met an agreement on the decision. Also, all the papers and data extracted from the papers were stored in the Google sheet for analysis by independent researchers.

4 Systematic Literature Mapping Results

In this part, we give the results of our proposed SLM study. We use "Pi" notation, referring to the primary study listed in Appendix 1.

4.1 RQ1. Background Understanding

With RQ 1, we focused on the definition of failure factors in the articles, looking for the trend of this context by year/avg. citation numbers, research facets, contributions facets, and the distributions of articles made by academia, industry or collaboration that is leading to the research on ERP failure factor.

4.1.1 RQ1.1. What is the trend of number of studies on ERP failure factors?

In this section, we focused on the publication frequency by year and examined the average citation characteristics of the articles. In Figure 3, we represented the distributions of articles with respect to years. According to the results, we observed that articles examining the ERP failure factors have emerged from 1999 to 2020, and there has been an increasing trend between 2009-2019 on this research subject. However, the number of articles that examined ERP failure factors decreased. We may conclude that increasing the successful ERP software implementations cause the decrease of the articles focusing ERP failure factors.

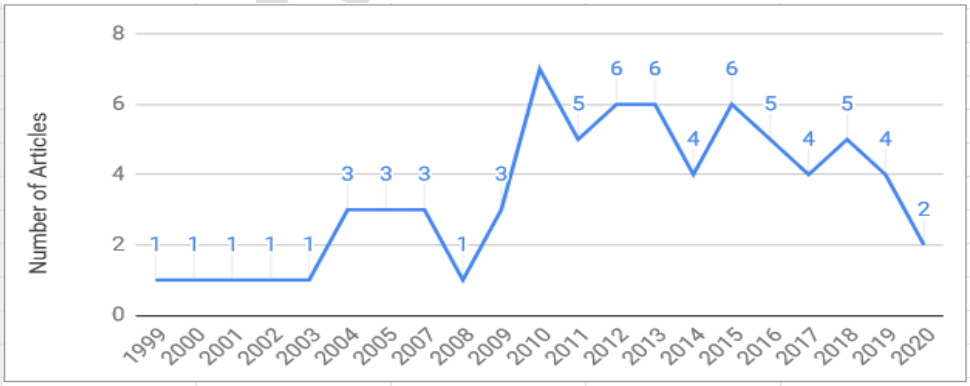


Figure 3. Distributions of articles per year

In order to see the mostly cited articles in the context of ERP failure factors between 1999-2020, we represented the visualization of several citations and an average number of citations per year in Figure 4. On the left side of the Figure, we represented citation count vs year, and on the right side of the Figure, we represented normalized citation count vs year. Normalized citation count formulized as shown below:

$$\text{Normalized citation count} = Total\ citation\ count/(2021 - publication\ year)$$

According to the results, we can see that while the top-cited article is the study of Gargeya and Brady [42] with a 579 citation count, the top-cited article according to normalized citation count belongs to Calogera [43].

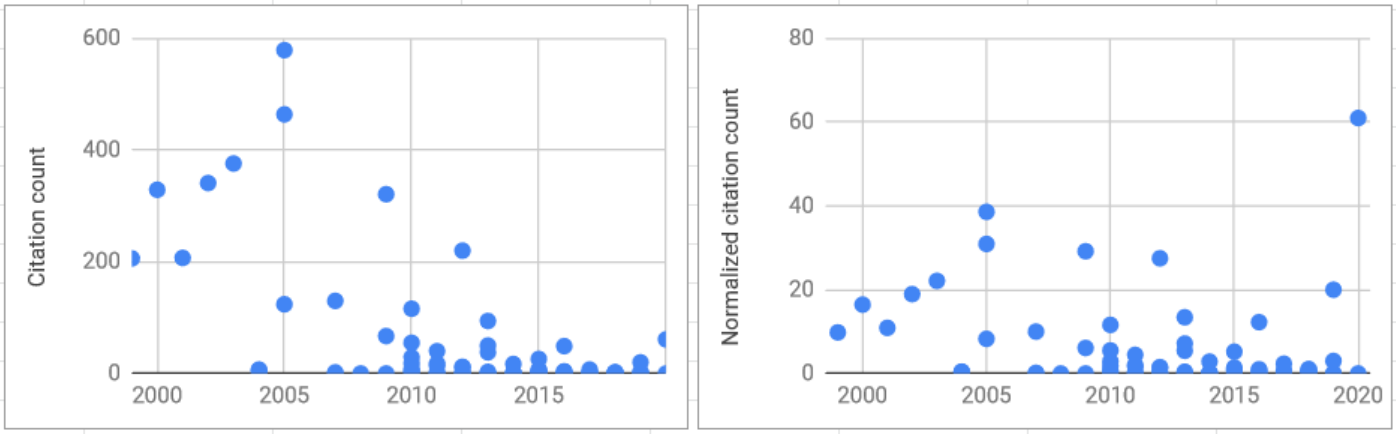
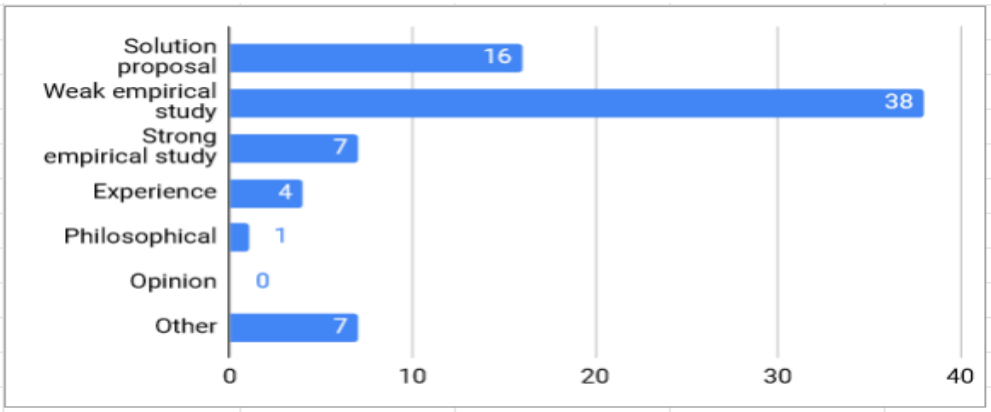


Figure 4. Citation counts vs. publication years of the studies

4.1.2 RQ1.2. What are the research methods developed in the studies?

According to Petersen guideline [41], we extracted data with respect to solution proposals, weak empirical study, strong empirical study, experience papers, philosophical papers, and opinion papers. To understand the maturity of this field, we represented the distributions of research methods developed in the studies, as shown in Figure 5. While the research made on ERP failure factor was developed mainly by weak empirical studies (38 articles, %0.52), the solution proposal research method followed it by 16 articles. The fact that there are very few strong empirical studies in the literature, indicating that this field is actually not very mature.

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Solution Proposal: P3, P12, P19, P20, P25, P27, P32, P47, P54, P57, P61, P63-P65, P67, P69

Weak Empirical Study: P1, P2, P4, P6, P8, P9, P13, P15, P17, P18, P21-P24, P28-P31, P34, P35, P37-P43, P45, P48-P50, P52, P53, P55, P56, P68, P71, P72

Strong Empirical Study: P5, P7, P16, P26, P33, P46, P59

Experience Paper: P10, P11, P60, P66

Philosophical Paper: P36

Other: P14, P44, P46, P51, P58, P62, P70

Figure 5. Distribution of research methods employed in the studies

4.1.3 RQ1.3. Who is leading the research in ERP Failure Factors (Academia, Industry, Collaboration)?

In Figure 6, we gave the analysis of the results by academic, industry or collaboration between these two. According to the results, it was determined that 15 articles were conducted with the academy, 9 studies within the industry, and 5 studies were conducted with academia-industry cooperation. The results show that the interest in software quality in the field of artificial intelligence-based software is higher in the academy area than in the industry.

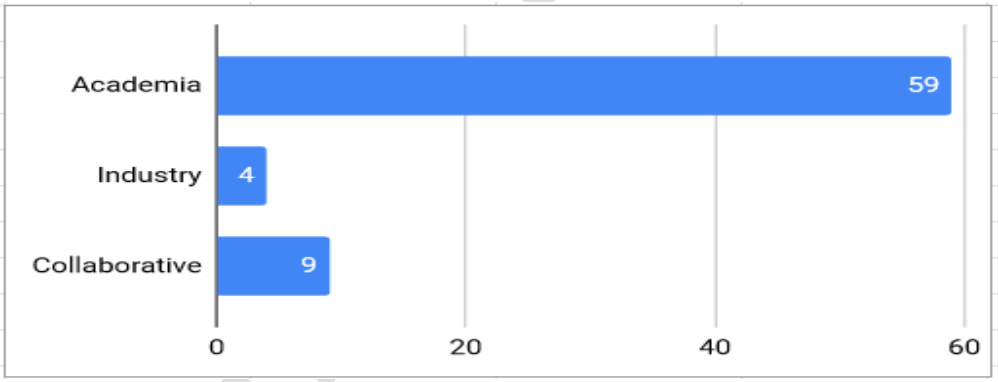


Figure 6. Distribution of author affiliations in the studies

4.1.4 RQ1.4. What type of contributions have been made in this area up to now (model, metric, tool, case study, etc.)?

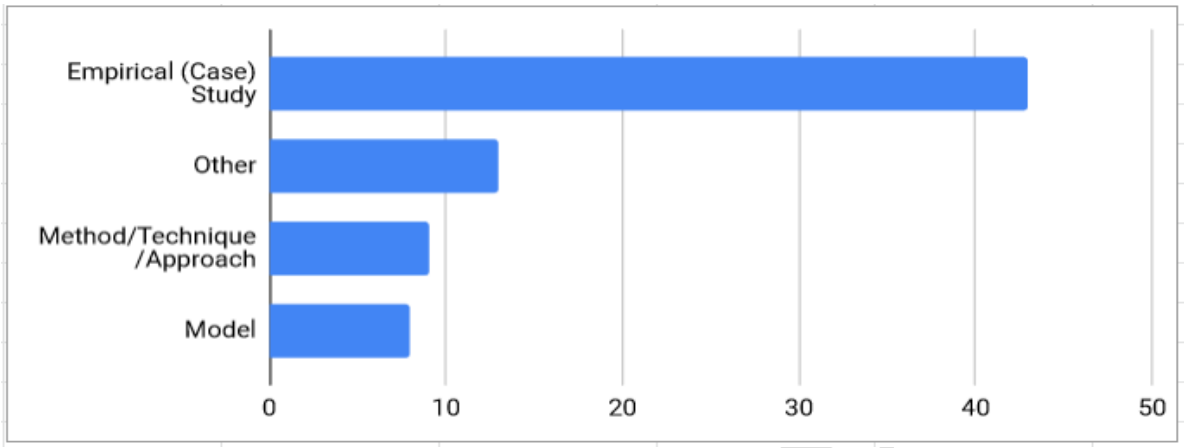
With this question, we analyzed the studies according to their contribution facets. In Empirical (**Case) Study**: P1, P2, P5-P9, P11, P13, P15, P18, P19, P21, P24-P26, P28-P30, P32, P34-P37, P39-P41, P44, P45, P50, P52, P55, P58-P62, P64, P66, P68, P71, P72

Other: P10, P12, P16, P20, P31, P33, P38, P42, P51, P54, P57, P69, P70

Method/Technique/Approach: P14, P27, P43, P46, P47, P48, P53, P65, P67

Model: P3, P4, P17, P22, P23, P49, P56, P63

Figure 7, the distribution of contribution facets by studies was represented. According to the results, while 43 articles contributed to an empirical (case) study, there were 13 articles that we could not put into under any categorization column. These articles were in general review studies, guidelines, etc.



Empirical (Case) Study: P1, P2, P5-P9, P11, P13, P15, P18, P19, P21, P24-P26, P28-P30, P32, P34-P37, P39-P41, P44, P45, P50, P52, P55, P58-P62, P64, P66, P68, P71, P72

Other: P10, P12, P16, P20, P31, P33, P38, P42, P51, P54, P57, P69, P70

Method/Technique/Approach: P14, P27, P43, P46, P47, P48, P53, P65, P67

Model: P3, P4, P17, P22, P23, P49, P56, P63

Figure 7. Distribution of contribution facets by studies

4.2 RQ2. Qualitative Coding in ERP Implementation Projects

4.2.1 RQ2.1. How failure factors are defined in ERP?

In order to understand the researchers' perspectives on ERP failure factors, we proposed to extract the definition of these failure factors identified in the articles.

Within the scope of this study, a total of 806 CFFs were identified in 72 articles examining the causes of failure in ERP projects. The full list of 806 CFFs can be viewed in Appendix 2. When this list is examined carefully, it will be seen that very different words are used for failure factors that have approximately the same meaning. For example, Tsai [44] defines it as "Reach" for the project team to interact with end users to understand the requirements, while Malaurent [45] defines it as "communication matters". In different articles, the exact words are used with different meanings. For example, according to Sahran [46], "Lack of internal communication" is seen as a subset of "Change Management".

Correct classification is very important for a good analysis using 806 CFF extracted from 72 articles. The qualitative coding approach has been carefully applied for this accurate classification because there is no doubt that the qualitative data coding process is an important part of this systematic literature mapping.

Qualitative coding can be described as creating classifications from the interpretation of data [47]. During this study, inductive and deductive approaches were combined. First of all, we did the first-round pass at coding. After this stage, our codes were organized into categories. After that, we did further rounds of coding. In the last step, we turned the codes into our final 17 categories. Word clouds related to these seventeen factors can be seen in Figure 8. Also, Figure 8.s shows the word cloud created by 806 CFFs. This word cloud proves how wrong it is to see ERP projects as IT projects. In other words, ERP projects should be considered as business and management projects that affect all processes and ways of doing business.

a. Project Management Problems b. Lack of Top Management Support c. Lack of Training



d. Poor Project Team



e. Poor Organizational Culture



f. Lack of Clarity on Scope



g. ERP Software Misfit



h. Poor Solution Implementer



i. Communication Problems



j. Poor quality of BPR



k. Financial Management Problems



l. Highly Customized ERP



m. Resistance to change



n. Outdated IT infrastructure



o. Poor Quality of Testing



p. Lack of User Involvement



r. Data Quality



s. Critical Failure Factors







Figure 8. The word clouds of failure factors

4.2.2 RQ2.2. How failure factors are categorized?

In the article pool of this study, there are 24 articles that examine articles from different perspectives by categorizing them. A complete list of CFFs and their categorization can be found in Appendix 3. 13 of these 24 articles that categorize CFFs used the titles "Organization" and "Technology". This is followed by the "Personal" category with 9 articles. The main categories of failure factors can be seen in Figure 9.

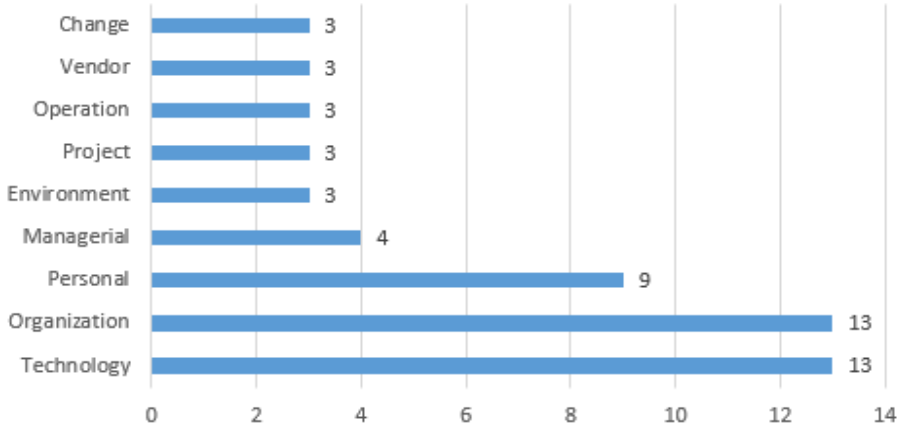


Figure 9. The main categories of failure factors

Many articles use traditional categories such as "Organization" and "Technology"; Tsai [44] sees the earth orbiting the moon as a great metaphor for an ERP system. With this approach, Tsai classified the failure factors as centripetal forces and centrifugal forces. Khanna [48], on the other hand, classified the failure factors as functional and non-functional. From a different perspective, Thiak [49] has made its classification as cost/time and customization. Original classifications of failure factors in articles can be seen in Table 4.

Article	Original Classification Schema
P01	Change Management
P02	Organization, Personal issue
P05	Employee/Personnel/HR & Process related, End-user Related, Enterprise related, Performance related, Project related, Strategy related, Technology & Vendor related
P06	Environmental Context, Organizational Environment, Technology Context
P07	Centrifugal Forces, Centripetal Forces
P08	Cost, Operational, People, Strategic, Tactical, Technology
P09	Functional, Non-Functional
P17	Human resources, Managerial, Organization, Technical, Vendors and Consultants
P23	ERP post-implementation Failure, Lack of external supports, Managerial, Operational, Technical, Top management, User's inefficiency
P30	Change Management, Customer Mindset, Process Flow, Software
P31	Cost and time, Customization
P34	Culture, Environment, Technical issues
P36	Macro level (organization), Micro level (individuals)
P37	Managerial, Technical
P42	Human, Organization, Technology
P46	Human resources, Managerial, Organizational, Processes, Project Management, Technical, Vendor and Consultant
P47	Customer related, Vendor related
P48	Characteristic of the project, Organization characteristics, Organizational learning
P55	Knowledge Challenges, Organizational Challenges
P56	Actor, Change, Fit, Plan
P57	Architecture and design, Implementation, Operational, Organizational
P63	Climate for implementation, Innovation-values fit
P67	Human and Organization, Technical
P72	IT experts, System, Top Management

Table 4. Original classifications of failure factors in articles

4.2.3 RQ2.3. Which stage of the ERP implementation process do we encounter the failure factor? (Pre-Implementation, Implementation, Post-Implementation)

Only 5 of the 72 articles in the pool of this study associated failure factors with the steps of the ERP project. The full list of the relationship of the failure factors with the steps in the ERP project can be seen in Appendix 4. When these 5 articles are analyzed, failure factors can be divided into 3 steps as Pre-Implementation, Implementation and Post-Implementation. While failure factors such as "ERP Software Misfit", "Lack of Top Management Support", and "Lack of Training" are important for each step, factors such as "Outdated IT infrastructure" can only be important in "Implementation" and "Post-Implementation" steps. The relationship of failure factors with project steps can be seen in Table 5.

Pre-Implementation	Implementation	Post-Implementation
Communication Problems	ERP Software Misfit	ERP Software Misfit
Data Quality	Highly Customized ERP	Lack of Clarity on Scope
ERP Software Misfit	Lack of Clarity on Scope	Lack of Top Management Support
Financial Management Problems	Lack of Top Management Support	Lack of Training
Lack of Clarity on Scope	Lack of Training	Lack of User Involvement
Lack of Top Management Support	Outdated IT infrastructure	Outdated IT infrastructure
Lack of Training	Poor Organizational Culture	Poor Organizational Culture
Poor Organizational Culture	Poor Quality of BPR	Poor Project Team
Poor Project Team	Poor Quality of Testing	Project Management Problems
Poor Quality of BPR		Resistance to Change
Poor Solution Implementer Performance		
Project Management Problems		
Resistance to Change		

Table 5. Relationship of failure factors with project steps.

4.2.4 RQ2.4. Which ERP failure factor causes what kind of failure of the ERP implementation project? (Process Failure, Expectation Failure, etc.)

Failure in ERP Projects does not occur in a single way. There are many different effects and modes of failure. According to the data extracted from the articles in the pool of this study, there are 12 different failure effects and 4 different failure modes. Ravasan [50] and Shirouyehzad [51], identified the modes of failures and stated which factors caused what mode of failure. Ravasan [50] revealed 4 different modes of failure, namely "process failure", "expectation failure", "interaction failure" and "correspondence failure". As can be seen in Table 6, for example, "Outdated IT infrastructure" causes "Interaction Failure", while "Lack of Training" causes "Expectation Failure" and Correspondence Failure. Shirouyehzad [51] presents the FMEA approach, which is used to identify the potential failure effects and control failure factors that affect ERP implementation. For example, "Lack of User Involvement" causes "Customer & Employee Dissatisfaction". The full list of failure factors and their main effects on ERP projects can be seen in Appendix 5.

Type	Potential Failure
PFM*	**Process Failure:** Poor Solution Implementer Performance, Lack of Top Management Support, Communication Problems
PFM	**Expectation Failure:** Lack of Training, Poor Solution Implementer Perf, Communication Problems
PFM	**Interaction Failure:** Communication Problems, Lack of Top Management Support, Outdated IT infrastructure
PFM	**Correspondence Failure:** Lack of Training, Lack of Top Management Support, Poor Solution Implementer Perf
PFE*	**Budget Exceed:** Poor Solution Implementer Perf, Lack of Top Management Support, Communication Problems
PFE	**Time Exceed:** Poor Solution Implementer Perf, Lack of Top Management Support, Communication Problems
PFE	**Project Stop:** Communication Problems, Poor Solution Implementer Perf, Lack of Top Management Support
PFE	**Poor Business Performance:** Lack of Training, Poor Solution Implementer Perf, Lack of Top Management Support
PFE	**Inadequate System Reliability and Stability:** Highly Customized ERP, Outdated IT infrastructure, Poor Solution Implementer Perf
PFE	**Low Organization Process Fitting:** Lack of Training, Poor Solution Implementer Perf, Communication Problems
PFE	**Low User friendliness:** Poor Solution Implementer Perf, Lack of Top Management Support, Communication Problems
PFE	**Low Degree of Integration and Flexibility:** Poor Solution Implementer Perf, Outdated IT infrastructure, Highly Customized ERP
PFE	**Low Strategic Goals Fitting:** Communication Problems, Poor Organizational Culture, Poor Solution Implementer Perf
PFE	**Bad Financial Performance Organization:** Communication Problems, Lack of Top Management Support, Poor Organizational Culture
PFE	**Cost increasing:** Financial Management Problems, Lack of Top Management Support, Communication Problems
PFE	**Customer & Employee Dissatisfaction:** Poor Project Team, Lack of User Involvement, Lack of Training
	* PFM: Potential Failure Mode
	** PFE: Potential Failure Effect

Table 6. Relationship of failure factors with failure modes and effects

4.3 RQ3. Ranking of ERP Failure Factors

4.3.1 RQ3.1. How is the order of importance of ERP failure factors determined?

20 of the 72 articles in the pool prioritized the failure factors according to their order of importance. The common purpose of these articles, which use various prioritization methods such as "survey for ranking", "occurrences in the projects", and "prioritized failure factors", is to prioritize the factors that cause failure. Some articles have multiple rankings for different

reasons. For example, P19 made two different rankings for two different projects. P65, on the other hand, made different sequences for different failure types. The full list of detailed rankings is available in Appendix 6.

In Table 7, the top five most important failure factors in each article can be seen in order of importance. While the number "5" is the most important failure factor, the number "1" is the 5th most important failure factor. Failure factors with the same order of importance are given the same order number. For example, in article P05, both "Lack of Top Management Support" and "Lack of Clarity on Scope" have the same order of importance.

"Project Management Problems" draws attention as the most important failure factor in ERP projects. It should be explained very clearly here that ERP projects are specialized projects with long-term and serious project management needs. It should not be confused that many factors such as "budget", "communication", and "project team" are not examined in the "Project Management Problems" category since they are examined in separate categories because many articles analyzed in this study show that project management is handled as project coordination. Therefore, unrealistic project scheduling, implementation strategy, and change management which is generally mentioned in some articles, are included in project management.

	P03	P04	P05	P08	P09	P09	P15	P18	P19	P19	P33	P37	P38	P40	P42	P49	P53	P56	P59	P65	P65	P65	P69	P71	T
Project Management Problems			1	1	1			3	3		5	4	5	1	4		4	1	4	3			5		**45**
Lack of Top Management Support		4	3	2	4			1	5	3						3		3	1	4	5			4	**42**
Poor Project Team	1			5	3		5		4								5	5	1	2		5			**36**
Poor Organizational Culture		1							2	4	2	2		1	1		3			5	3	3	1		**28**
Lack of Training		3											3	1	5			4	5				4	1	**26**
ERP Software Misfit			1			4	3	1					2								3	4	1		**19**
Poor Quality of Testing			5			1	3	1		5									3					1	**19**
Communication Problems	5	2				5				1				1					2	1	1				**18**
Resistance to change	3		1	3	5		1					2			2		1								**18**
Lack of Clarity on Scope			3			3		1						1		3		2	1				3		**17**
Poor quality of BPR	4							3	1	2	1		4											1	**16**
Financial Management Problems	1				2							5			3		2				1				**14**
Poor Solution Implementer Perf…		5	1					3				1		1					1						**12**
Lack of User Involvement				4			1				4											1			**10**
Highly Customized ERP						2					3													4	**9**
Data Quality													1			3									**4**
Outdated IT infrastructure																						1			**1**

Table 7. Ranking of failure factors

"Project Management Problems" are followed by "Lack of Top Management Support", "Poor Project Team", "Poor Organizational Culture", "Lack of Training" in order of importance. "ERP Software Misfit", which includes subheadings such as "ERP software selection", is only important in the 6th place. These data reveal that ERP projects should not be considered as an ordinary software project.

Poor top management support has been identified as one of the important factors that lead to the failure of ERP projects. How the top management approves the ERP project is very important. Bringing the necessity of the project to the table by the management and convincing the management of the project by others mean very different situations. Management forced into such a large project may view the ERP project as a secondary job. In particular, he may not adequately allocate his valuable human resources to the project. You should know that the success of the projects in which the management is reluctant will not be high. As the project manager, you must take the necessary measures immediately. You should inform the management, especially when you have a shortage of resources. Only 20%-30% of ERP projects are related to IT. Everything else is about interpersonal relationships and dedication to the project. If you cannot use critical resources in the later stages of the project, your project will not be finished on time, and even if you do, it will not be of the desired quality. At the beginning of the project, you should clearly indicate how much time the project leaders and end users will devote to this project. If you cannot guarantee that this resource will be allocated, you will ensure the failure of the project. The number one reason for booming projects is that the top management does not understand the project and does not provide adequate support. There are failures where limited financial support contributed to a rushed ERP implementation process project. This resulted in the team members being overloaded which resulted in a high staff turnover rate, ineffective knowledge transfer, and political problems occurred which hindered the implementation process. Management's commitment to quality is very important because they provide the resources for implementation and maintenance of the management system which contains all the processes related to quality.

The weakness or strength of the project team can be shown as the third most important factor in ERP projects. Building a strong project team is very important. Typically, an ERP implementation team includes an executive sponsor, a project manager, and representatives of key business groups involved in the project. Having an executive sponsor who can adjust business priorities and attract additional resources as needed is vital to success. Responsibilities of the team; It includes setting high-level goals, creating requirements and key performance indicators (KPIs), ensuring the project stays on time and within budget, daily project management, and measuring results. Because of the expertise required by project management functions, companies with fewer resources may need the help of an outside consultant to meet these challenges. During implementation, the team should be able to resolve conflicts and mediate between different groups within the organization. It may also need to make decisions about midstream changes in ERP implementation strategy based on user feedback and input. Therefore, team members must be both knowledgeable and highly respected within the organization.

4.3.2 RQ3.2. What failure factors and how often are encountered in failed ERP implementation projects?

With this question, we aimed to see the distribution of failure factors in each year interval by dividing the period between 1999-2020 into five-year periods as shown in Figure 10.

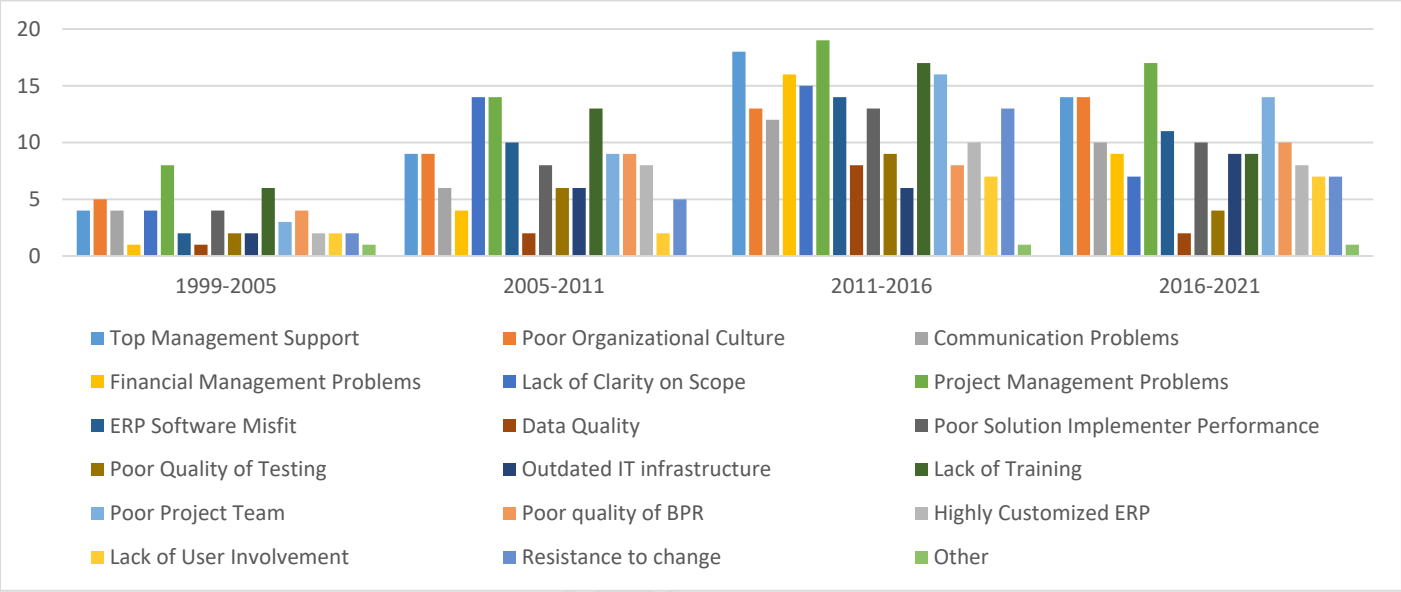


Figure 10. Distribution of failure factors into five-years periods

According to results, more attempts on failure factors in ERP projects were performed in the years between 2011-2016 and there is an increasing trend after 2010 with the context of ERP failure factors. While "Project Management Problems" mostly studied factor within all year periods, "Lack of Training" was the second most studied factor until 2016. After 2016, "Top Management Support", "Poor Organizational Culture", and "Poor Project Team" had the same and second mostly encountered factors with 14 articles.

During the adaptation of ERP projects, companies need to make various changes and regulations in their business processes. While these activities are carried out, the organizational culture of the companies will have an impact on these changes and regulations. Therefore, the impact of organizational culture on ERP projects is inevitable. There has been some increase in Culture over the last 5 years. The main reason for this expected result is the abandonment of the concept of adapting ERP to culture. In order for ERP to be successful, it is to ensure the transformation of culture without making many changes on the defined best practices. In recent years, when this understanding has gained importance, it seems normal to speak more of culture as a factor of failure.

There has been a significant increase in Outdated IT Infrastructure in the last five years. Thanks to successful ERP projects, ERP has gained importance for organizations. Over time, ERP has become indispensable for organizations. However, the renewal of ERP is becoming increasingly important in order to improve not only technological changes but also business processes. However, considering other failure factors, ERP renewal is not easy at all. This causes Outdated IT Infrastructure to be a much more visible failure factor.

ERP projects were seen as software projects in the first years and tried to integrate software into existing processes. This has either brought failures or brought much more significant costs. Studies in the last 5 years show that organizations that do not adapt to best practices and care about BPR are more likely to face failure.

4.4 RQ4. ERP Failure Factors for Failed ERP Projects

4.4.1 RQ4.1. How many different companies, projects and people have been worked with while determining the ERP failure factors?

The 45 articles in the pool collected data from one or more organizations. In this study, there are data from 369 organizations in total. Only 18 of the articles in the pool stated how many different ERP projects they collected data from. In this systematic literature mapping, data were collected from a total of 265 different ERP projects. Finally, 19 articles addressed how many different personnel they interviewed in organizations to extract failure factors. Data were collected through questionnaires and interviews with a total of 1090 personnel who have different responsibilities in the organization.

4.4.2 RQ4.2. Which geographic region's data was used when extracting ERP failure factors?

In order to get an overview of the most active region in this context, we analyzed the data used in the articles according to geographic regions. First, we analyzed the countries to which the data belongs according to the related continents as shown in Figure 11. There are seven categories as a continent. In the Asia category, there were thirty-three articles and mostly observed countries in this region were China, India, and Iran. In the America category, there were nine articles that belong to the USA, Canada, and Brazil. In the All category, there were seven articles, and in these articles, the authors specified the countries as global. In the Europe category, there were six articles, and they used data belongs to the UK, the UN, Romania, Norway, and Ireland. There were just two articles that used the data belongs to Middle Eastern countries. It is observed that while it is observed that the data belonging to the countries in the Asian continent is used at the highest rate with 46%, this is followed by the NA category with 15%, that is, the category of articles that do not indicate which country the data belongs to.

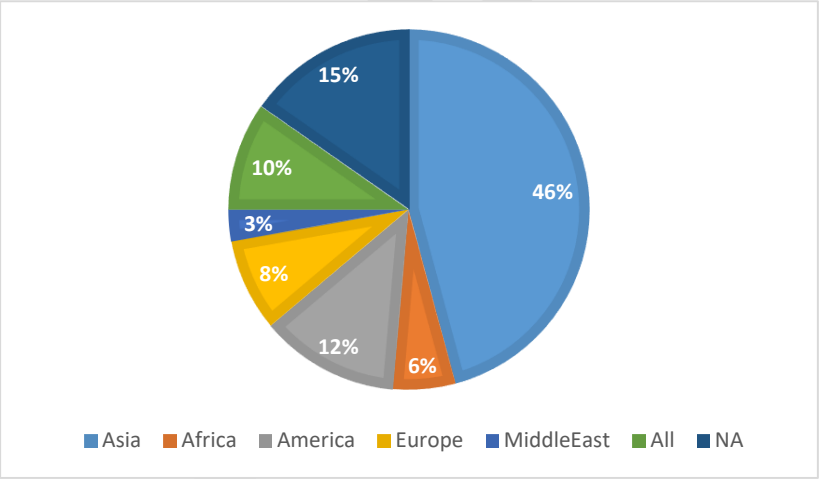


Figure 11. Distribution of articles according to the continents to which the data belongs

Secondly, we made observations on which failure factor was used more than which continent as shown in Figure 12. According to results, in Asia, while mostly studied failure factor was “Project Management Problems” with twenty-three articles, “Poor Organizational Culture” followed it with twenty-two articles. In America, Europe, All, and NA category, the articles mostly focused on “Project Management Problems” same with Asia. Even if there are very few articles in the Middle East, the failure factor “Poor Solution Implementer Performance” was the most studied factor in this region.

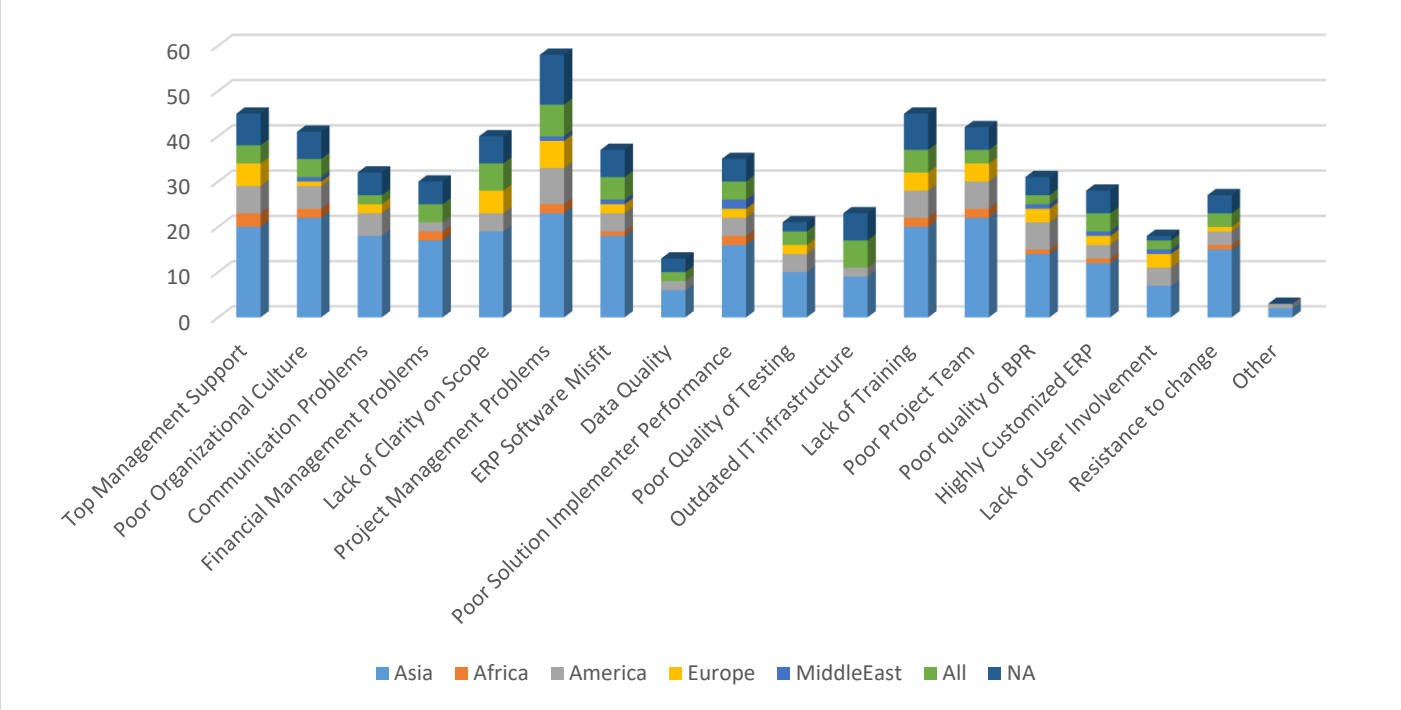
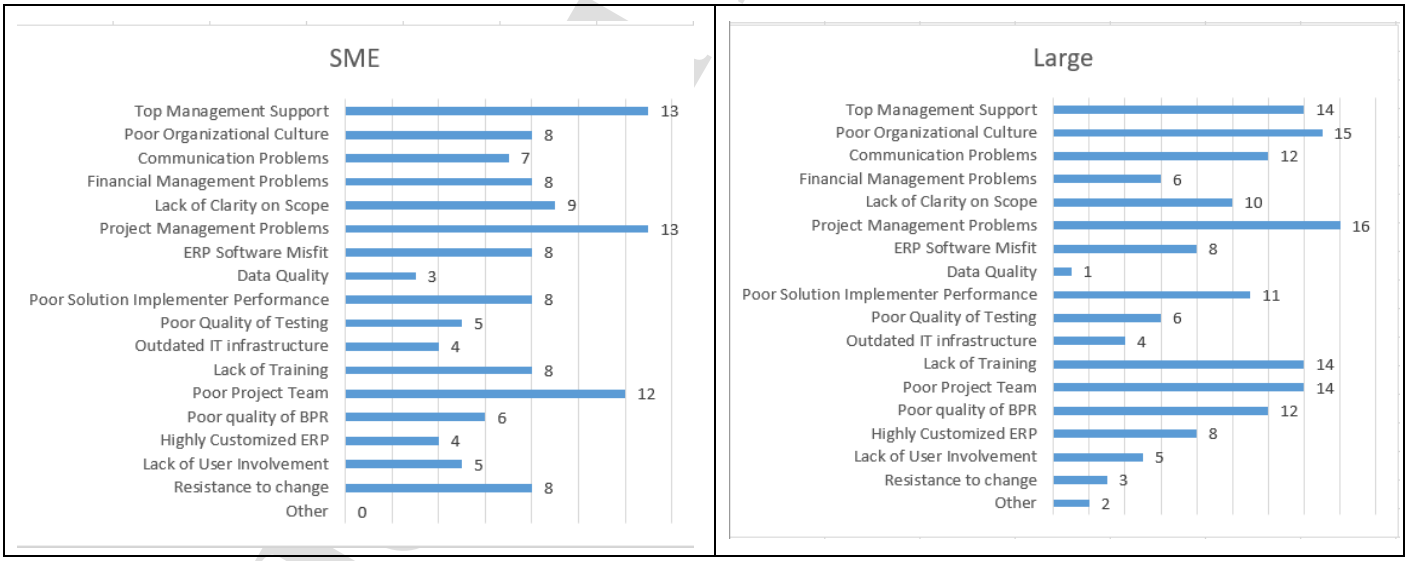


Figure 12. Distributions of failure factors according to continents

4.4.3 RQ4.3. What size companies' data were used when extracting ERP failure factors? (SME, large, all, etc.) factors?

With this question, we propose to understand the frequency of failure factors with respect to the size of the company data. We categorized the size as SME (small-medium enterprise), large, all, and NA (Not assigned). In Figure 13, we can see the distribution of failure factors with respect to the size of companies. If we look at the figure, we can see that in all sizes of the companies, the "Project Management Problems" factor was mostly considered a factor among all failure factors. In addition, in large companies, "Poor Organizational Culture" is the second most studied factor.



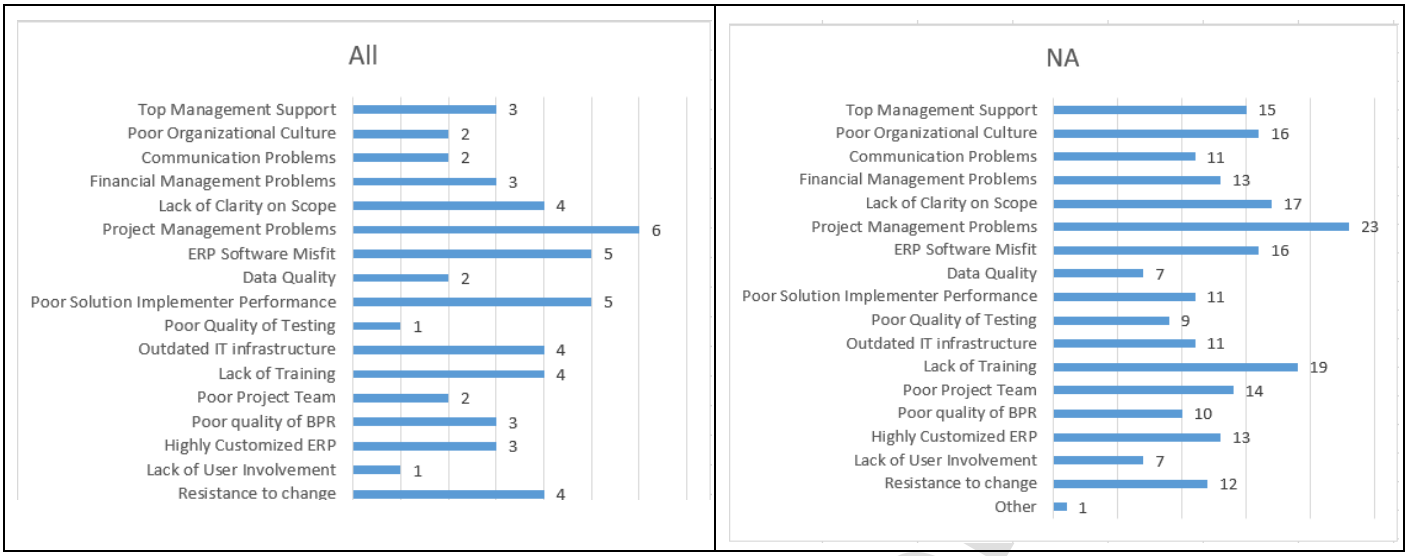


Figure 13. Distribution of failure factors with respect to size of companies' data

Project management problems are mostly encountered problems in both SME and large-scale companies. A bad practice that both small and large-scale companies do in project management is to resort to practices such as changing the software and developing new interfaces in some approaches of a system that are incompatible with the company's processes. In particular, some standardized packages require the company to adapt its own processes to the software according to the requirements of the software. It is inevitable that companies make their own adaptations to the software instead of adapting some of their processes to this system, and it will come back to itself as a big problem in version upgrade studies.

The second most common problem in ERP projects is top management support. The lack of ownership of the project by the top management, the mis-implementation of the project according to the wishes of different departments without focusing on the main objectives, and the lack of sufficient seriousness and importance lead these projects to failure.

As expected, education, organizational culture, and communication problems come to the fore in large-scale companies, while financial problems come to the fore in small-scale companies. Communication problems between departments are striking in large-scale companies. ERP projects fail as communication becomes inefficient. Likewise, education budgets are either very insufficient or cannot be used efficiently. This does not change for ERP either. Since the examination, standardization, and dissemination of processes in large-scale companies are within the framework of the information systems environment, the concept of individuality is much less common with the minus of institutionalism. This causes resistance to change to be encountered relatively less. However, in small-scale companies, unwritten processes, the fact that every job has a hero, and the heroes think that their importance will decrease with the ERP change, this problem comes to the fore. In addition, ERP projects are an important opportunity to review and improve existing business processes. Most companies believe that they should transfer the existing business style to the ERP solution. While large companies are aware that if they could manage their processes with 100% performance and success, they would not need such a solution anyway, while small-scale companies may lag behind in this regard. All ERP solutions are developed with best practices and management styles in mind. That's why large-scale companies believe that managing certain processes as the software suggests and not resisting change often pays off.

Every company, large or small, that installs ERP systems faces the cost and complexity of that system, but the worst problems are It is about setting up ERP systems without fully understanding and thinking about all business concepts. The installation of ERP systems, in addition to wasting money and time, can disrupt the culture of the company, create a wide training requirement and even lead to reduced production and customer orders that cannot be evaluated. For many small-scale companies, financial problems affect investment and growth. Those who can manage these problems properly can embark on large-scale journeys. As a matter of fact, problems may arise in allocating a budget for ERP or in the use of the allocated budgets.

4.4.4 RQ4.4. What type of industry's data were used to extract ERP failure factors?

According to our experience and the research we have done in the literature, ERP systems are not specific to any type of industry, they are for the use of all industries. When the articles in the pool of this study are examined, 23 articles do not mention which type of industry they contain, while only 5 articles indicate that the failure factors are valid for all types of industry. As can be seen in Non-profit org.**:** P71

Logistic: P56, P64

Telecom: P19, P53

Retail: P08, P35

Automotive: P09, P17

University / Education: P13, P45, P51

Professional Services: P22, P39, P51

Financial / insurance: P02, P51, P53, P56

High-tech / IT: P01, P39, P52, P60, P63

All: P37, P38, P49, P57, P59

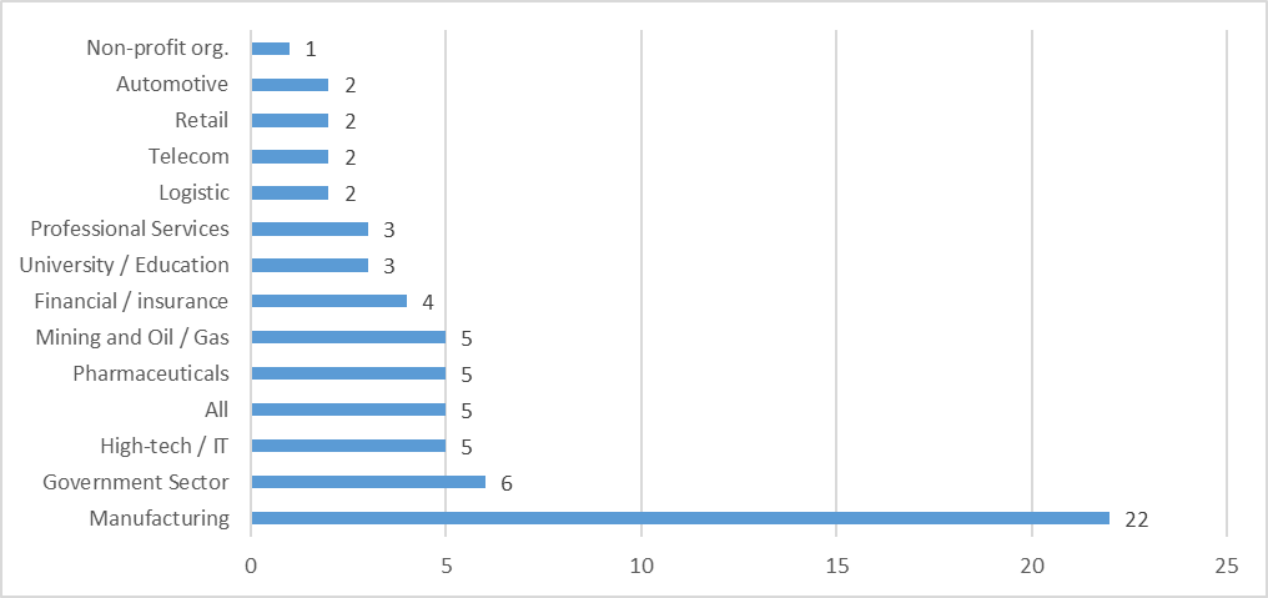
Pharmaceuticals: P09, P34, P48, P51, P66

Mining and Oil / Gas: P15, P24, P33, P36, P53

Government Sector: P19, P24, P28, P33, P55, P61

Manufacturing: P01, P06, P07, P09, P11, P18, P22, P23, P29, P30, P32, P34, P40, P42, P53, P56, P62, P63, P65, P67, P69, P72

Figure 14, the area of industry where the failure factors are most extracted is "Manufacturing". This is followed by the Government sector.



Non-profit org.: P71

Logistic: P56, P64

Telecom: P19, P53

Retail: P08, P35

Automotive: P09, P17

University / Education: P13, P45, P51

Professional Services: P22, P39, P51

Financial / insurance: P02, P51, P53, P56

High-tech / IT: P01, P39, P52, P60, P63

All: P37, P38, P49, P57, P59

Pharmaceuticals: P09, P34, P48, P51, P66

Mining and Oil / Gas: P15, P24, P33, P36, P53

Government Sector: P19, P24, P28, P33, P55, P61

Manufacturing: P01, P06, P07, P09, P11, P18, P22, P23, P29, P30, P32, P34, P40, P42, P53, P56, P62, P63, P65, P67, P69, P72

Figure 14. Types of industry

4.4.5 RQ4.5. How is data gathered in order to derive critical failure factors (interview, questionnaire, literature review, etc.)?

In this part, we proposed to extract the data sources such as interviews, personal experiences, literature reviews, archived records/documents, surveys, and others (opinions, observations, etc.) to derive the critical failure factors in the articles. In Interview: P2, P6, P9, P15, P18, P23, P25, P28, P31, P33-P35, P39, P42, P50, P52, P53,

P55, P56, P61, P63, P64, P68 P71, P72

Personal Experience: P1, P3, P10, P11, P13, P21, P24, P29, P30, P32, P36, P39, P40, P43, P44,

P46-P49, P51, P57, P60, P62, P66, P71

Literature Review: P4, P5, P8, P12, P14, P19, P20, P23, P27, P33, P37-P40, P43, P46,

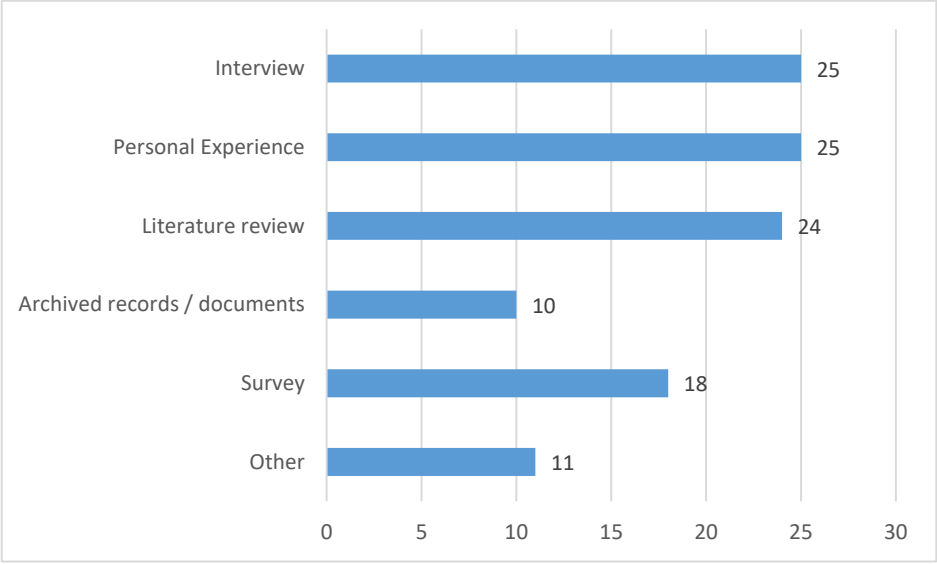
P49, P51, P54, P57, P59, P65, P67, P69

Archived records/documents: P2, P6, P15, P28, P34, P39, P50, P56, P67, P68

Survey: P5, P7-P9, P16, P17, P22, P23, P26, P28, P33, P37, P42, P45, P46, P50, P63, P69

Other: P6, P9, P26, P28, P34, P37, P41, P44, P50, P67, P70

Figure 15, we represented the distribution of each data source, and the results showed that the top three data sources used in the articles were interviews, literature reviews, and personal experiences.



Interview: P2, P6, P9, P15, P18, P23, P25, P28, P31, P33-P35, P39, P42, P50, P52, P53,

P55, P56, P61, P63, P64, P68 P71, P72

Personal Experience: P1, P3, P10, P11, P13, P21, P24, P29, P30, P32, P36, P39, P40, P43, P44, P46-P49, P51, P57, P60, P62, P66, P71

Literature Review: P4, P5, P8, P12, P14, P19, P20, P23, P27, P33, P37-P40, P43, P46, P49, P51, P54, P57, P59, P65, P67, P69

Archived records/documents: P2, P6, P15, P28, P34, P39, P50, P56, P67, P68

Survey: P5, P7-P9, P16, P17, P22, P23, P26, P28, P33, P37, P42, P45, P46, P50, P63, P69

Other: P6, P9, P26, P28, P34, P37, P41, P44, P50, P67, P70

Figure 15. Data sources to derive critical failure factors

5 DISCUSSION

5.1 BENEFITS OF THIS STUDY

This study, ERP implementation failure systematic literature mapping, set sight on ERP implementation project managers, consultants, organizations, and researchers. ERP implementation project managers and consultants can take advantage of this study while they implement their ERP projects in organizations. This study is the most comprehensive research in the literature in terms of the identification and classification of ERP failure factors. ERP project managers can use this comprehensive research and reduce the risk of failure. Moreover, using this information, organizations can better interpret the data of the ERP implementation project.

Additionally, solution implementers can use all this information and results to perform their consultancy more efficiently. Consultants can improve their communication and solution implementation process with all this information. Also, researchers can extend this mapping study with their research. They can extend result questions and exhibit new research.

Some researchers consider ERP projects as standard software projects. However, this is a wrong assessment. This survey has revealed in many ways that ERP projects are not ordinary software projects. For example, "ERP Software Misfit", which includes subheadings such as "ERP software selection", is only important in the 6th place. However, the most important factors such as "Project Management Problems", "Lack of Top Management Support", and "Poor Organizational Culture" are related to project management, and these are completely specific to ERP. As this study shows, it is crucial not to view ERP projects as standard software projects to avoid failures.

806 items were detected in 72 articles on failure factors. The use of so many different names in the literature makes it difficult to perform the analysis correctly. A qualitative coding process has been adopted to overcome this difficulty with unstructured or semi-structured data. In this way, the most comprehensive process of systematically categorizing the critical failure factors was implemented. In this respect, we think that this study will be an essential guide for future studies.

Lack of Top Management Support comes to the fore in this study as one of the most critical factors leading to the failure of ERP projects. Limiting top management support to providing the necessary resources for the project will lead to the failure of the project. Because, within the scope of the ERP project, top management support is essential for both process and organizational changes to be accepted in all units of the organization. In addition, the implementation of the project according to the wishes of different departments without focusing on the main objectives will bring failure. Therefore, it is vital to manage the strategic goal correctly.

It is unrealistic to expect employees to be immediately proficient in the new ERP system. Targeted, continuing education that meets the needs of different groups and roles can help users accept and get the most out of the system. One approach suggests providing customized content such as videos and tutorials and allowing employees and others to select the most relevant content for their jobs. It is also important to provide hands-on training to help employees learn about the system. Some organizations have found that the best practice is to select some users for early, comprehensive training and use them to guide others by sharing their experiences and skills. Therefore, lack of training is also the other important factor for ERP project failures.

It takes a long time for ERP to be used and become widespread over time. It is clearly seen that while the failure factors, which were mentioned much more in the past, lost their popularity over time, some failure factors gained more importance over time. For example, since the importance of education has been understood, it draws attention as a failure factor that has been studied less in the last five years. On the contrary, business process improvement has gained momentum over the past five years. This study is also of great importance in terms of revealing trends.

The size of the organization can change the failure factor that needs attention. As a matter of fact, severe differences can be seen between CFFs depending on the organization's size. For example, resistance to change and financial problems may be encountered more commonly in small-scale organizations, whereas communication is more critical in large-scale companies. Therefore, with this study, organizations of different sizes can focus on the failure factors that are more important to them.

From this perspective, as expected, most of the articles are directly related to the manufacturing industry. However, it is known that the use of ERP is becoming widespread in other sectors as well. This study showed that gaps can be found in areas outside the manufacturing industry regarding ERP failure factors. New studies for different types of industries will be made in the coming years.

There are a very limited number of articles giving information about the relationship between failure factors and ERP implementation stages, although there can be different failure factors that need to be focused on in each phase of the project. Because ERP projects are long-term, multi-phase projects. In this study, all the articles in the pool were carefully reviewed, and it turns out that some failure factors such as "ERP Software Misfit", "Lack of Top Management Support", and "Lack of Training" are important for each step. These results show that there is a strong relationship between the failure factors that should be considered in each phase of the project and the failure factors that are considered important.

It is very important to correlate failure factors with failure types. Because, even if the failure factor that causes the failure type cannot be improved, alternative improvements that can make something up of that failure factor can be made much more effectively and at a lower cost. Within the scope of our study, we have made a very detailed examination on this subject. However, our results showed that there is a significant gap in the literature on this issue.

5.2 THREATS TO VALIDITY

The most important validation threats in such systematic literature mapping studies can be listed as follows: reliability of data extraction, missing article pool due to insufficient search terms, selection of article databases, correct definition, and application of exclusion/inclusion criteria. These validation threats are discussed based on a standard checklist adopted from (Claes et al., 2012).

Internal validity: The strategy used for source selection is discussed in detail in Section 3.2. The purpose of this strategic approach is to ensure that this systematic literature mapping research is reproducible. With this approach, search terms and engines, as well as inclusion and exclusion criteria, are defined and recorded in full detail. In this context, an incomplete definition of search terms and engines is one of the biggest internal validity risks. In addition, the careless application of inclusion and exclusion criteria by the authors is another major internal validity risk.

Inadequate search engines may cause deficiencies in the created article pool. In addition, to create a complete article pool, search terms must also be determined completely. Different authors have independently developed their own search terms on different search engines. Then these search engines and words were reviewed and consolidated, and then a systematic search was performed. After all, the references of the articles included in the pool were also examined to further reduce the risk of missing related articles.

Another internal validity risk is that the inclusion and exclusion criteria may be interpreted differently by the authors for various reasons. In order to prevent this threat and to create a high-quality article pool, a comprehensive and systematic peer review process has been applied.

Construct validity: In our study, special attention was paid to the research questions in order to ensure construct validity. The authors determined the research questions by discussing them in detail. In addition, categorization schemes were developed using previously used and proven classification schemes to eliminate construct validity risks.

Conclusion validity: This sort of validity of SLM is related with the getting of appropriate conclusions by means of strict and repeatable analysis. Each paper within the pool was assessed by means of least two authors to decrease mistakes for data extraction. Contrasts of assessment between the authors were settled with the agreement. Following standard

procedure provided repeatability of this mapping study. Moreover, this process guaranteed that the results of similar research will not have critical divergences from our findings.

External validity: In such systematic literature mapping studies, external validity can be defined as the validity of applying the results of the research outside the context of the study. Data were extracted from all academic and industrial articles related to ERP Failure. These data were turned into sufficient information for industry and academia. The most important point to remember here is that the results in this research focused only on lessons learned in failed ERP projects. We are not inclined to extend our findings beyond this scope.

6 Conclusion and future work

In this study, we presented a systematic literature mapping in order to analyze the state of the art in the failure factors of ERP projects and distinguish what we have about this area. Our intent is to benefit practitioners and analysts by providing the most comprehensive research in the field of critical failure factors of ERP.

In this systematic literature mapping, a very comprehensive pool of articles was created from studies investigating why ERP projects fail. This paper mainly examines types of failure factors, common names of failure factors, the main categories of failure factors, the relationship of failure factors with project steps, and failure modes/effects on a systematic literature mapping including 72 papers. In addition, this article investigated the trend of failure factors, the variation of critical failure factors according to the area/size of the organization, and the effect of the organization's geolocation on failure factors. With the developing technologies, ERP is a growing research and application area which is not only for industrial companies but also for all organizations that want to use their resources efficiently. We recommend that all organizations that use or will use ERP consider the information generated from this comprehensive analysis study. In addition, we have seen with our study that; there are many studies on the importance of failure factors. However, the number of studies on the effects of these factors on each other and how to reduce these negative effects are very few. Lastly, the lack of studies on the systematic measurement of failure factors draws attention.

The direction of our future work is to develop industry-academy collaboration by utilizing the results of this systematic literature mapping. In addition, our future research includes studies that can systematically measure the value of each failure factor. The studies that model the relationships of these failure factors with each other and the ability to control these failure factors will be the continuation of this study.

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